LectureLens

Offline lecture search with links back to the source

Snapdragon AI Lab Build & Present Challenge

The problem

A useful explanation can be buried inside a long recording.

Students need the moment the lecturer spoke and the slide shown at that point.
A transcript alone makes those sources hard to connect.

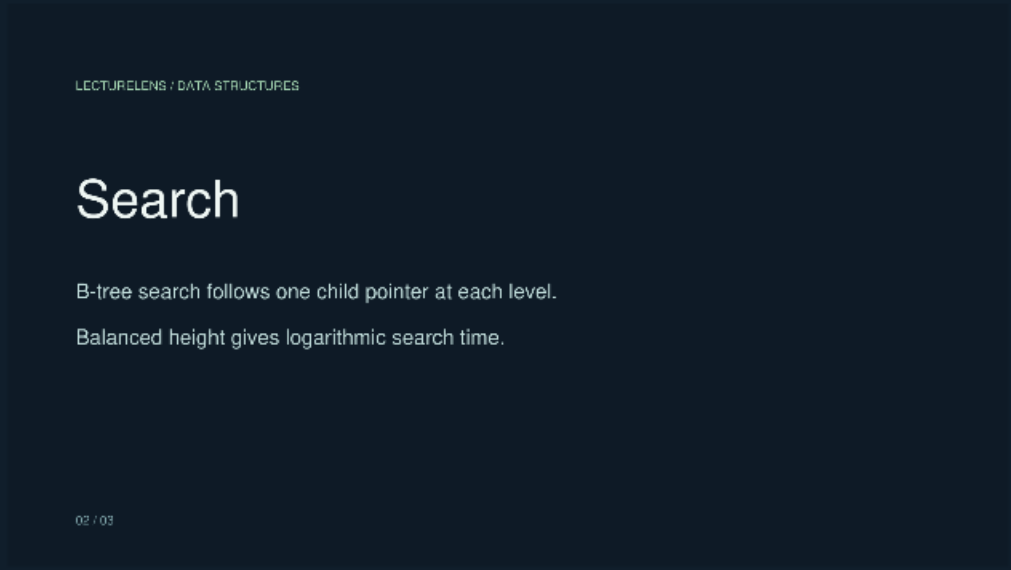
LectureLens keeps both sources searchable in one local timeline.

A search result opens the source

1 Import a recording, transcript, and slides

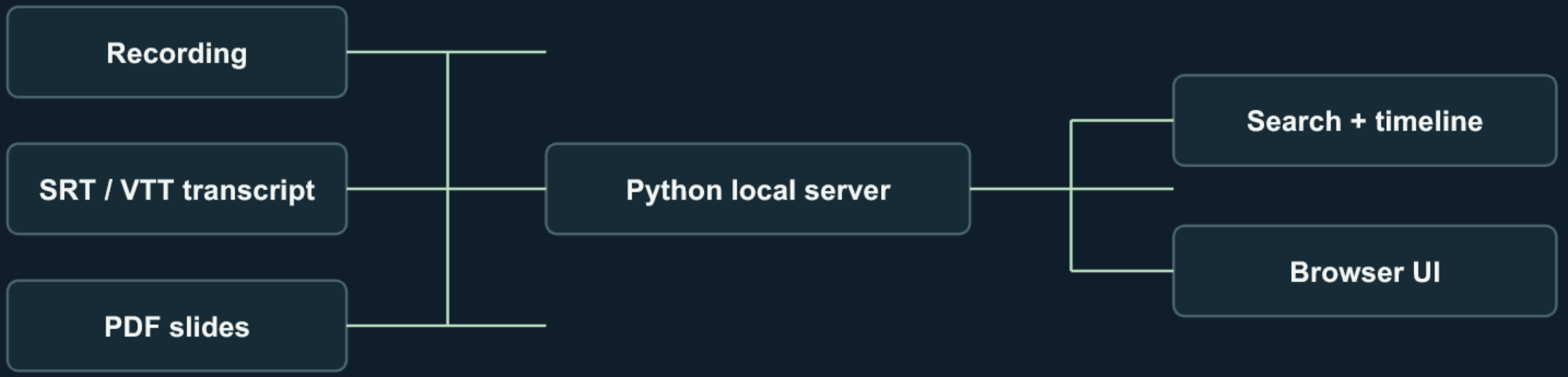
2 Search a concept across spoken and slide text

3 Open the audio time and linked page



Sample: “logarithmic” returns the transcript at 0:18 and slide 2.

Local architecture



Optional Qualcomm Whisper adapter for WAV transcription

NPU execution needs validation on the target laptop

Prototype status

Working today

Local import and storage
PDF text extraction and slide preview
Timestamped transcript search
Audio seek and manual slide links

Still to validate

Qualcomm Whisper assets on the target PC
Actual QNN execution provider
Latency and battery measurements
ARM64 installer

Snapdragon validation plan

Target: Windows 11 on a Snapdragon-powered HP PC

1. Run Qualcomm’s Whisper sample with matching AI Hub assets
2. Confirm the actual execution provider on the device
3. Compare transcription time on the same audio clip
4. Package and document the ARM64 app after validation